

ELCT 302

Laboratory Two:

Model Identification

Group 2
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I hereby certify that I have complied
with the Spirit and the Letter
of the Carolinian Creed
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Abstract

This lab report will be discussing the processes that our group went through to complete the objectives of this portion of the project. The introduction will explain what is completed in the lab, the theory section will talk about all of the concepts that serve as the foundation for this project. In the procedure, we will talk about the specific tasks that we accomplished and the methods that we used to do so. We will also have a results section that will show the measurements taken as well as the values and functions that we calculated.

Table of Contents

Table of Contents.....	2
I Introduction	3
II Theory	3
A. <i>DC Motor Theory</i>	<i>3</i>
B. <i>Transfer Function.....</i>	<i>3</i>
C. <i>Transfer Function Estimation.....</i>	<i>5</i>
D. <i>Ackerman Steering</i>	<i>5</i>
E. <i>Bicycle Model.....</i>	<i>6</i>
III Experimental or Procedure.....	7
A. <i>Servo Motor Model Identification</i>	<i>7</i>
B. <i>Dynamic Steering Frame</i>	<i>8</i>
IV Discussion / Results.....	9
A. <i>Transfer Function Estimation</i>	<i>9</i>
B. <i>Dynamic Steering Frame</i>	<i>12</i>
V Conclusions.....	13
A. <i>Servo Motor Model Identification</i>	<i>13</i>
B. <i>Dynamic Steering Frame</i>	<i>13</i>
VI Appendix.....	14
A. <i>Microcontroller Code</i>	<i>14</i>
B. <i>MATLAB Code for Servo Motor Model Identification</i>	<i>15</i>
C. <i>MATLAB Code for Dynamic Steering Frame</i>	<i>15</i>
D. <i>Center of Mass and Moment of Inertia Calculation</i>	<i>16</i>
VII References	17

Laboratory Two

I Introduction

One of the objectives of this lab was to design a circuit that would perform a step response of the servo motor. In addition, this data was used in transfer function estimation. The step response design, as well as the transfer function design both used important topics in model identification. Another important task in this lab that used model identification was estimating the dynamic steering frame. This estimation also took key ideas from Ackerman Steering, and the bicycle model. The objectives and tasks completed during this lab are important to the control and precise steering of the final car.

II Theory

A. DC Motor Theory

DC motors were designed in order to convert electrical energy to mechanical energy. There are three key parts of the DC motor, which are the stator, the armature, and the commutator [2]. The stator is a set of non-moving magnets, on which the winding is wound. It also has cavities between its poles. The armature is a rotating part, consisting of a cylinder of insulated magnetic laminations. The commutator is a set of copper segments, stacked together to make a cylinder, each one insulated from the last [3]. Each of these key components are labeled in the figure below.

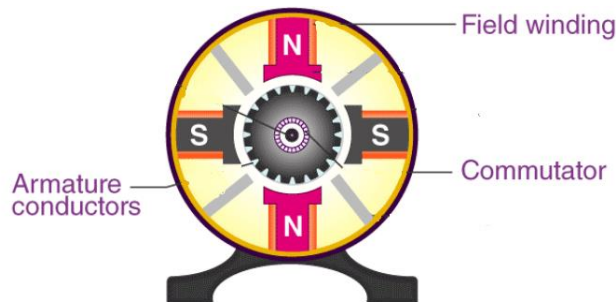


Figure 1: DC Motor Parts Diagram

The magnetic field created from the stator is what causes the rotation of the armature. The previously mentioned winding on the stator is connected to another key component, called the commutator. This commutator both applies an electrical current to the windings, and also allows the armature coils to create torque, and as a result rotation [2]. This mechanical energy is the needed output of the DC motor.

B. Transfer Function

The transfer functions are important to find because they can help us in building a model of our system. This model can help us get a better understanding of the workings of the system.

The first part of this lab focuses on estimating a transfer function for our servo motor. It is important for us to first identify the type of servo motor we are utilizing. Our servo motor is a DC Armature controlled servo motor. With that established, we now have to find the equation for the KVL loop within the motor armature. This can be seen in (Eq. 1). This equation is found by making the sum of voltage increases and drops around the loop of the armature equal to zero. For all “ $e_a(t)$ ” is the applied voltage, “ i_a ” is the armature current, “ R_a ” is the armature resistance, “ L_a ” is the armature inductance, and “ $e_b(t)$ ” is the back emf. Next, we need to find the equation for the moment of inertia. This equation is shown in (Eq. 2). It is found by setting the sum of forces of the various torques on the motor equal to zero. “ $T_m(t)$ ” is the motor torque, “ $T_L(t)$ ” is the load torque, “ $\theta_m(t)$ ” is the angular displacement, “ $\omega_m(t)$ ” is the angular velocity, “ J_m ” is the motor inertia, and “ B_m ” is the viscous friction. We have to find the Laplace transform of these state equations to get the transfer function. “The transfer function can be defined as the ratio of the Laplace transform (LT) of the o/p variable to the LT of the i/p variable. Generally, the DC motor changes the energy from electrical to mechanical. The supplied electrical energy at the armature terminals is changed into controlled mechanical energy [5].”

$$e_a = i_a R_a + L_a \frac{di_a}{dt} + e_b(t) \quad \text{Eq. 1}$$

$$J_m \ddot{\theta}_m = T_m - T_L - B_m \omega_m \quad \text{Eq. 2}$$

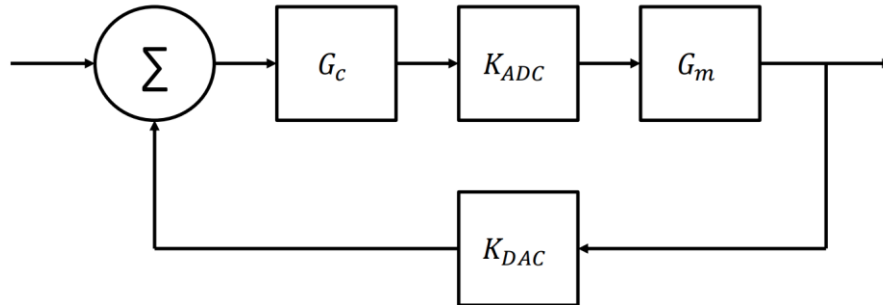


Figure 2: DC Armature Controlled Servo Motor Block Diagram

As we discussed in the transfer function estimation section, we had to figure out which controller gain (G_c) we needed to use in order to most accurately represent our system. Shown above is the block diagram for the servo motor. G_m is the basic transfer function (Eq. 3), which is multiplied by the controller gain (G_c). We can use block diagram reduction to find our final transfer function (Eq. 4). As briefly mentioned before, there are three controller models that could be accurate. We chose to use the proportional integral derivative (PID) model to calculate the transfer function because this was most accurate to the data we gathered, and we replaced G_c with the equation for the PID model. Again, this is discussed further in the estimation section. Using the PID model, the final transfer function is shown in (Eq 5). This transfer function has 2 zeros and 4 poles.

$$G_m = \frac{1}{s} \left(\frac{K}{JL_a s^2 + (JR_a + BL_a)s + R_a B + K^2} \right) \quad \text{Eq. 3}$$

$$\frac{G_c(s)K_{DAC}G_m(s)}{1 + G_c(s)K_{DAC}G_m(s)K_{ADC}} \quad \text{Eq. 4}$$

$$\text{Transfer Function} = \frac{K_p s + K_i + K_d s^2}{s^4 + s^3 + (1 + K_d)s^2 + K_p s + K_i} \quad \text{Eq. 5}$$

C. Transfer Function Estimation

The three possible transfer functions correspond with the three types of possible controllers that are being used in the servo motor [8]. The controller can either be proportional (P), proportional integral (PI), or proportional integral derivative (PID) [8]. The general transfer function can be seen in Equation 4. While the value of G_m is not dependent on the controller, G_c value is. This G_c value affects the number of zeros and poles in the transfer function in Equation 4. The value for each is shown below.

$$G_c = K_p + \frac{K_i}{s} + K_d s \quad \text{Eq. 6}$$

$$G_c = K_p + \frac{K_i}{s} \quad \text{Eq. 7}$$

$$G_c = K_p \quad \text{Eq. 8}$$

The equations above show the values of G_c for the PID (Eq. 6), PI (Eq. 7), and P motor (Eq. 8). The first step in finding the transfer functions is to find the transfer function of a PID motor. The G_c value from Equation 6 is used to solve Equation 4. From the equations above, you can see that from this first transfer function, K_d can be set to zero, and then both K_d and K_i equal to zero in order to solve for the transfer functions of the PI and P motors.

D. Ackerman Steering

Ackerman Steering is a mechanism of a type of steering that allows the front wheels of a vehicle to be properly angled in order to turn efficiently. Turning efficiently in vehicles, like automobiles, are important to prevent slipping and allow for full stability of the vehicle. One of the main points of Ackerman Steering is the instantaneous center of curvature (ICC) or “the common centre of the turning radius” which when both of the wheels are turned, they always are pointed towards the ICC, which can be seen in Figure 3 [6]. The red lines are from the ICC to the center axles of the wheels, which are perpendicular, though the wheels have different angles. So, while steering, the wheel on the inside of the steer is at a sharper angle than the outside wheel of the turn. This is due to the inside wheels requiring a shorter distance traveled compared to the outside wheel. The sharper angle on the inside wheel prevents it from sliding on the ground while turning, which makes it ideal.

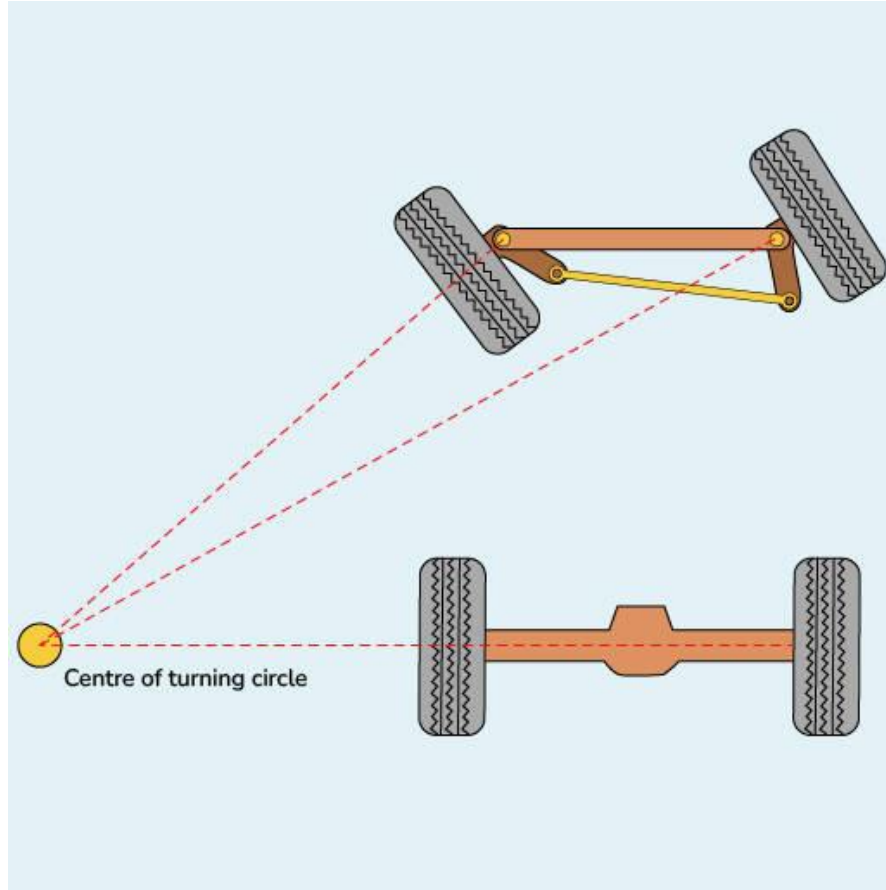


Figure 3: Diagram showing the front wheels aligned towards the ICC

E. Bicycle Model

A typical car vehicle has four wheels, but simplifying it into two wheels is what the bicycle model is all about. Like how the Ackerman Steering uses four wheels and involves two steering angles for the front wheels, the bicycle model only deals with one steering angle. The purpose of the bicycle model is to create a state-based model with state-space equations in order to describe it at any point in time, but there are important assumptions that are required. “The vehicle is operating in a 2D plane [...] the vehicle has a lumped mass all acting at the center of mass [...] the velocity of the wheel acts in the same direction that the wheel is facing” [7]. Figure 4 shows the bicycle model itself and looking at the model, the states are shown, Y and φ . Y is the distance from the center of mass (COM) to the ICC and φ is the angle of the vehicle heading. Figure 5 shows the given matrix of the state-space equations to solve for the y , y' , φ , and φ' . Given the general cornering stiffness, C , and the calculated variables of distance from COM to axles, L , rotational moment of inertia, I_z , velocity of the car, V_x , and the mass of the vehicle, m , are used and plugged into the equations onto MATLAB to solve. Finally, the limitations of the bicycle model is that it is an ideal model, thus not including lateral forces affecting the model, such as the ground surfaces and centrifugal force. Other limitations include no possible slipping of the tires, and the equations are linearized, but in reality it is nonlinear.

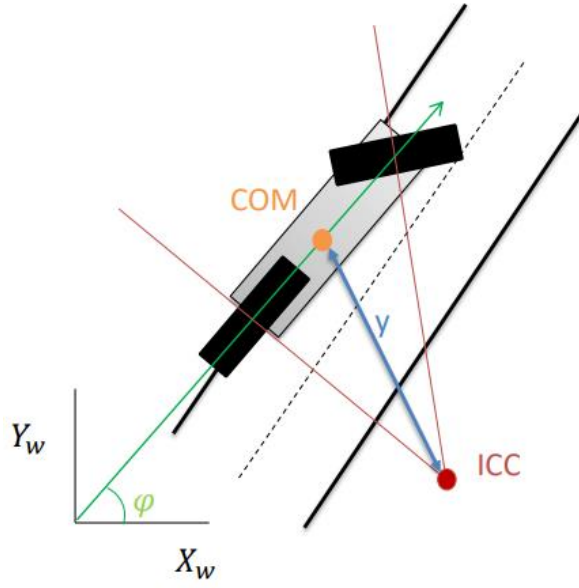


Figure 4: Bicycle Model

$$\frac{d}{dt} \begin{bmatrix} y \\ \dot{y} \\ \varphi \\ \dot{\varphi} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{2C_{\alpha f} + 2C_{\alpha r}}{mV_x} & 0 & -V_x - \frac{2l_f C_{\alpha f} - 2l_r C_{\alpha r}}{mV_x} \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{2l_f C_{\alpha f} - 2l_r C_{\alpha r}}{I_z V_x} & 0 & -\frac{2l_f^2 C_{\alpha f} + 2l_r^2 C_{\alpha r}}{I_z V_x} \end{bmatrix} \begin{bmatrix} y \\ \dot{y} \\ \varphi \\ \dot{\varphi} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{2C_{\alpha f}}{m} \\ 0 \\ \frac{2l_f C_{\alpha f}}{I_z} \end{bmatrix} \delta$$

Figure 5: State-Space Matrix Equations

III Experimental or Procedure

A. Servo Motor Model Identification

The first step in completing the bump test is to connect the components correctly. In place of the car's servo motors, a separate motor is given. This new motor is connected to the microcontroller, which will rotate the servo motor one radian using the given code in the Appendix. The motor is also connected to a potentiometer, which will be used to record the change in voltage.

The potentiometer has three pins, only one of which is the output. In order to determine which pin is the output, the multimeter is connected to any two pins, and the change in resistance is measured as the servo motor rotates one radian. This process is repeated for each different set of pins available. When the resistance does not change, this means that the third pin not connected to the multimeter is the output pin.

This output pin is connected to the oscilloscope, which will measure the change in voltage as the servo motor rotates. The given code in the Appendix will output a step function from the microcontroller. This output is equal to 1 when the servo motor is rotated, and 0 when it has not

rotated. The data from these two output signals is taken from the oscilloscope and put into an Excel spreadsheet. After being manually smoothed in Excel, it undergoes a smoothing function in MATLAB. The data from the step function will act as the input in the given MATLAB code, and the data from the potentiometer will be the output.

The number of zeros and poles in the three transfer functions can be estimated using the ideas mentioned in the *Transfer Function Estimation* theory section above. Once these results are input into the given MATLAB code, the transfer function itself can be estimated. From these transfer functions, the step response plots are produced. Each of these graphs are compared to the recorded step response in order to find the best match.

Another estimation is the Padé Approximation. This is based on the time delay taken from the original Excel spreadsheet. This involves one line of MATLAB code and will produce another estimation. The final transfer function chosen for the project can be either the best match previously mentioned, or this Padé Approximation.

B. Dynamic Steering Frame

The second week of the lab focuses on the physical and dynamics of the steering and the entire car's frame. The first step was to measure the mass of the car, which a scale was used.

Next was finding the center of mass (COM) of the car which Equation 9 can be used, where M is the total mass, m_i is a specific mass of a force or object acting on the overall car and x_i is the distance of that specific mass at a specified origin. Having the starting point or the origin at the back of the car, the entire length of the car, the length of the rear wheel axel in respect to the origin and the length of the front wheel axel in respect to the origin were measured. Using the total mass of the car and multiplying it by half of the entire length of the car is one term of the summation in the equation. The mass of the rear wheel of the car and its corresponding distance from the origin would be the second term. The mass of the front wheel and its distance would be the third term of the summation. Finally, the total mass of each component will be added and used in the denominator.

$$x_{cm} = \frac{\sum_{i=1}^N m_i x_i}{M} \quad \text{Eq. 9}$$

The third step was finding the moment of inertia. Since the overall shape of the car is a rectangular prism, the moment of inertia can be found using Equation 10 because the shape is just simply a rectangular prism. The total mass, m , the depth, d , and the length, l , of the car can be used to plug into the equation to solve for the moment of inertia with respect to the z -axis.

$$I_z = \frac{m(d^2 + l^2)}{12} \quad \text{Eq. 10}$$

Finally, the last step is to use the calculated values and given constants from the lab instructions and PowerPoint to validate the model estimation parameters. Using the matrix from Figure 5 and using MATLAB to create the matrix and insert the required variables of the equations

to solve for the states, y , y prime, φ , and φ prime. The MATLAB code is included in the Appendix below and it graphs the step response of each of the states.

IV Discussion / Results

A. Transfer Function Estimation

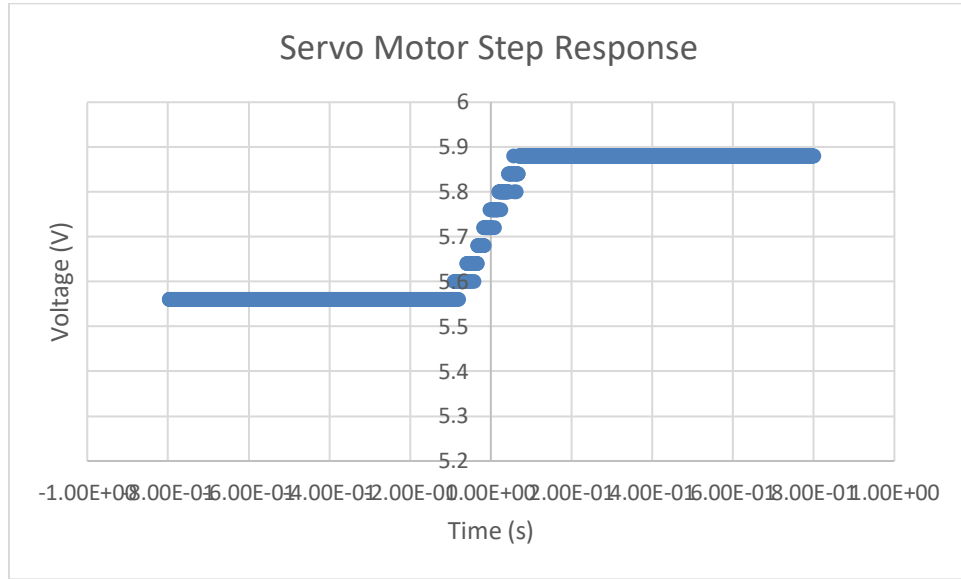


Figure 6: Recorded step response of servo motor

The above figure shows the step response recorded from the potentiometer that was connected to the servo motor. This response was uploaded to MATLAB, and also used to compare the three estimations of the transfer function in later steps.

The three transfer function estimations are below. Equation 11 is for the P controller, Equation 12 is for the PI controller, and Equation 13 is for the PID controller.

$$\frac{Y(s)}{R(s)} = \frac{-0.02291}{s^3 + 0.7618s^2 + 21.14s + 11.76} \quad \text{Eq. 11}$$

$$\frac{Y(s)}{R(s)} = \frac{1.135s + 64.25}{s^4 + 4.378s^3 + 13.47s^2 + 26.42s + 2.37e-6} \quad \text{Eq. 12}$$

$$\frac{Y(s)}{R(s)} = \frac{-6.239s^2 + 9.676s + 64.25}{s^4 + 6.048s^3 + 40.46s^2 + 81.07s + 39.91} \quad \text{Eq. 13}$$

From the equations above, it was determined that the transfer function for the P controller will have no zeros, and 2 poles. The transfer function for the PI controller will have 1 zero and 3 poles. The transfer function for the PID controller will have 2 zeros and 4 poles. These results are reflected in the given MATLAB code. Using this code, the step function was plotted for each of the above transfer functions. These plots are shown below.

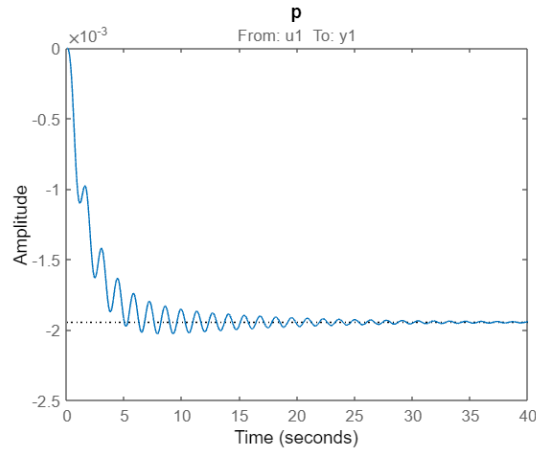


Figure 7: Step response of transfer function in Equation 11

Figure 7 shows the step function based off of Equation 11, which shows the response of the P controller. This was compared to Figure 6 and found to not be a close enough match. This means that the controller in the servo motor cannot be a P controller.

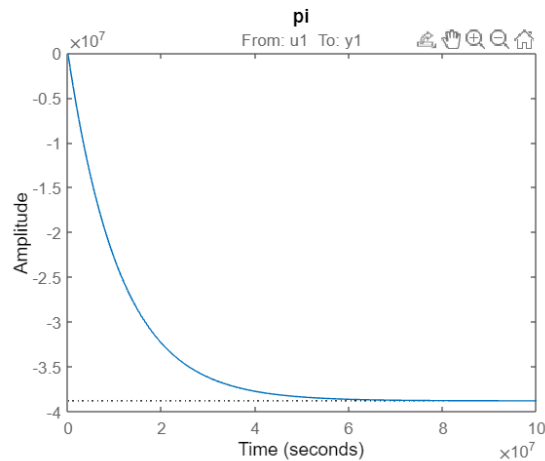


Figure 8: Step response of transfer function in Equation 12

The figure above shows the step response of the transfer function from Equation 12, which is the response estimated for a PI controller. This was compared to the recorded response in Figure 6, and as the general shape/slope does not match it was determined that the servo controller was not a PI controller.

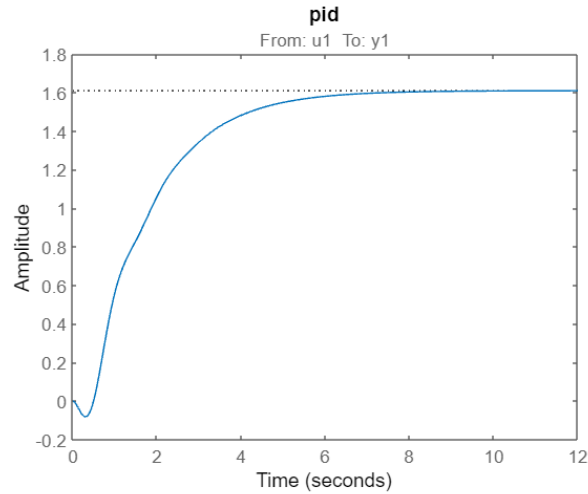


Figure 9: Step response of transfer function in Equation 13

Figure 9 shows the step function based off of Equation 13, which shows the response of the controller if it is PID. This was compared to the recorded step response in Figure 6 and found to be the best match. This means that the servo contains a PID controller, or a Proportional Integral Derivative controller.

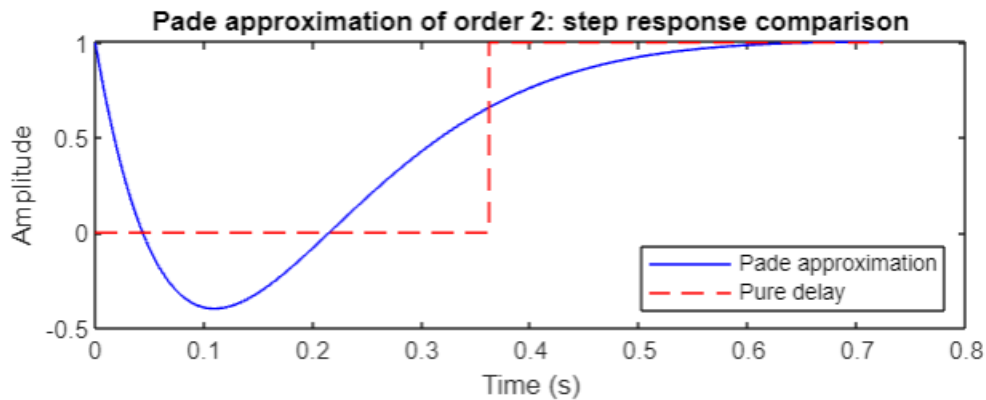


Figure 10: Padé Approximation of step response

This Padé Approximation shown above is a time delay approximation. It is found through the MATLAB command called 'pade()' that can be found in the given code. While this approximation is fairly accurate to the step response in Figure 6, it is not as correct as the estimation in Figure 9. This estimation is only accurate for a unit test, and if the motor turns more than one radian, the time delay will be slightly different, which will cause the above estimation to be slightly off.

B. Dynamic Steering Frame

Total Mass = 0.538 kg	Total Length = 0.2794 m
Front Wheel Axel Mass = 0.59 kg	Front Wheel Axel Distance = 0.2159 m
Back Wheel Axel Mass = 0.288 kg	Back Wheel Axel Distance = 0.0254 m
COM = 0.118 m	$I_z = 0.004676 \text{ kgm}^2$
Cornering Stiffness, $C = 22.5 \text{ kN/rad}$	$V_x = 1 \text{ m/s}$
Depth of Car = 0.161925 m	

Table 1: Calculated and given Constants

Looking at Table 1, these are all the calculated constants, like the total mass, front and back wheel mass, COM, total length, depth, front and back wheel distance, I_z , and the given constants like the cornering stiffness being between 20 to 25 kN/rad, so 22.5 kN/rad were chosen for calculations and 1 m/s was chosen for the car's velocity. The calculation of the COM and moment of inertia can be found in the Appendix below.

After writing the MATLAB code and taking the step response of the output states, Figure 11 shows the plot of the step responses. The first output step response corresponds to the f , which is the distance of the COM to the ICC. This linear characteristic of the step response goes to infinity due to the cornering stiffness. The cornering stiffness in this case was inputted into MATLAB as a constant of 22.5 kN/rad, so continues to go at that rate. The third output step response corresponds to ϕ , and that too goes to infinity. The reason for that is that when the car is turning and drives forward at that turn, it will forever go in a circle. Starting at zero degrees, one full cycle becomes 360 degrees, and another circle would make it to be 720 degrees and so on. Plots two and four shows the step responses of f' and ϕ' and those converge to a finite number. This is due to them being the derivative to f and ϕ , and when taking the derivative of a linear function, it leads to a constant.

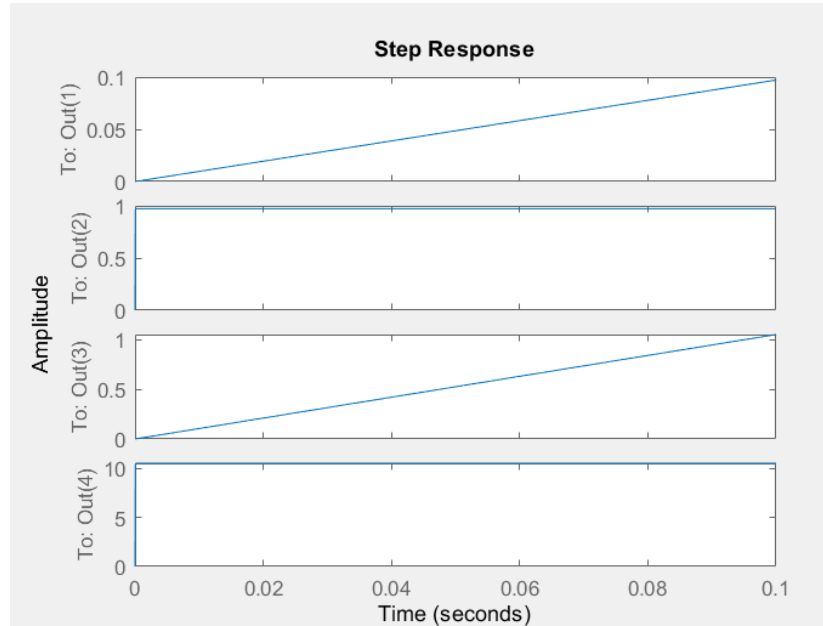


Figure 11: Step Responses of f , f' , ϕ , and ϕ'

V Conclusions

A. Servo Motor Model Identification

To conclude this report, all the of the requirements and objectives were met by the methods that we used within the project. We took measurements and gathered data with a potentiometer mounted to a servo motor to get a better idea of the impulse response of our servo. This allowed us to estimate a transfer function using MATLAB. We also completed some work on paper with the given transfer function to figure out which controller gain model most accurately represented the transfer function estimation that we got. We had to choose from PID, PI, or P models. We also completed a Padé approximation using MATLAB to see how it compared to the other results that we already found. This work to find the most accurate transfer function possible is very important in the process of modeling our system because it will help us in the future when we are doing further testing.

B. Dynamic Steering Frame

For this second part of the lab, we were tasked with modeling the steering characteristics of the frame of our vehicle, and again, all of the objectives and requirements were met. We began by measuring the mass of our vehicle. Then, we used a series of equations to calculate the center of mass based on the various forces acting on the frame of the vehicle. Next, we calculated the moment of inertia. After we had these calculated, we plugged our values in the state space equations and entered everything into MATLAB in order to estimate the radius of a path with a given input angle. This part of the lab, similar to the first part, will help us during further testing of our vehicle because we now have a better understanding of how the vehicle will react to certain inputs.

VI Appendix

A. Microcontroller Code

```
#include "mbed.h"
#include <stdio>
#include <iostream>

PwmOut PWM1(PTA1);
DigitalOut servoin(PTA2);
AnalogIn potvolt(PTB3);
float voltage;
float dutyCycle;
float zerorad;
float onerad;

int main()
{
    while(true)
    {
        zerorad = .05;
        onerad = 0.05/(90*3.14/180)+0.05;

        PWM1.period(0.02);

        PWM1.write(zerorad);
        servoin.write(1);

        printf("\nduty cycle: \n");
        printf("%f", zerorad);

        voltage = potvolt.read();
        printf("\nvoltage1: \n");
        printf("%f", voltage);

        wait_us(1000000);

        PWM1.write(onerad);
        servoin.write(0);

        printf("\nduty cycle: \n");
        printf("%f", onerad);

        voltage = potvolt.read();
        printf("\nvoltage2: \n");
        printf("%f", voltage);

        wait_us(1000000);
    }
}
```

B. MATLAB Code for Servo Motor Model Identification

```
output = readmatrix('2ouput.xlsx');
input = readmatrix('2input.xlsx');

out=smooth(output);

sys=iddata(out, input, 0.001);

figure;
step(tf1)
title('pid')

figure;
step(tf2)
title('pi')

figure;
step(tf3)
title('p')

%pid
tf1=tfest(sys,4,2)
%pi
tf2=tfest(sys,4,1)
%p
tf3=tfest(sys,3,0)
```

C. MATLAB Code for Dynamic Steering Frame

```
clc
clear all

Caf=22.5e3; Car=22.5e3; m=0.538; Vx=2; lf=0.0979; lr=0.0926; Iz=0.004676;

a = [0, 1, 0, 0; ...
      0, -(2*Caf+2*Car)/(m*Vx), 0, -Vx-(2*lf*Caf-2*lr*Car)/(m*Vx); ...
      0, 0, 0, 1; ...
      0, -(2*lf*Caf-2*lr*Car)/(Iz*Vx), 0, -
      (2*((lf)^2)*Caf+2*((lr)^2)*Car)/(Iz*Vx)];

b = [0; (2*Caf)/m; 0; (2*lf*Caf)/Iz];

c=eye(4);

d=zeros(4,1);

sys=ss(a,b,c,d)

step(sys)
```

D. Center of Mass and Moment of Inertia Calculation

$$x_{cm} = \frac{\sum_{i=1}^N m_i x_i}{M}$$

$$x_{cm} = \frac{(0.288kg * 0.0254m) + (0.538kg * 0.1397m) + (0.59kg * 0.2159m)}{0.59kg + 0.288kg + 0.538kg}$$

$$x_{cm} = 0.118 \text{ m}$$

$$I_z = \frac{m(d^2 + l^2)}{12}$$

$$I_z = \frac{0.538kg(0.161925m^2 + 0.2794m^2)}{12}$$

$$I_z = 0.004676 \text{ kgm}^2$$

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